

Unit 3 Prototype

What Is Prototyping?

Creating a physical version of a concept or experiment is a process known as prototyping.

The use of prototypes is crucial for product teams and designers. To build a fundamentally solid app or website, it's critical to visualize and test ideas early in the design-creation process. This also helps to reduce expenses while constructing the lifelike model you'll have before development. Prototype test it with potential customers and stakeholders to see whether it truly answers the problems of its consumers.

The objective is to give a system general functionality. **Prototyping in design thinking** has many benefits, including early error discovery, quicker user input, and a better knowledge of the system being produced for the users. Hence, prototyping is widely used.

Note :

- Prototyping is an essential step in the development of a product, be it digital or physical. The process converts the idea into a concept that has a physical structure. This concept can be presented to stakeholders or audience to test its viability and validate assumptions.

In simple terms, a prototype isn't just a part of the product design, it is one of the most integral parts to –

- Know if the ideated product is viable for mass production,
- Give an essence of the product to the stakeholders and investors before the actual product is developed.
- A prototype is a rudimentary working sample, model, mock-up or just a simulation of the actual product based on which the other forms (MVP, final product, and variations) are developed.
- The main motive behind prototyping is to validate the design of the actual product. Sometimes, creating a prototype is called materialization as it is the first step of transforming the virtual or conceptualized design into the real physical form.

Why do We Need to Prototype in Design Process?

One of the phases before releasing the finished product is building a prototype. Creating a product prototype has several benefits. Here are a few ideas for why prototyping is essential during the design process.

- Rapid prototyping design thinking enables its deployment by identifying the elements of an idea that are challenging or impossible to achieve. Making a prototype reveals technical and economic limitations that were not previously known.
- By analyzing the site's overall navigation, prototypes assist in testing the usability of a website.
- By gathering feedback from users in the form of reviews, recommendations, and testimonials, prototypes aid in the presentation of ideas and concepts to users in a more tangible way.
- As it immediately influences the most important aspects of a project, namely the resource, time, and budget, prototypes aid in minimizing risks, identifying covert flaws and functional gaps.
- Through their ability to entice potential customers to invest in them, prototypes help to simulate the ultimate product. It assists in evaluating the accuracy of a design and locating design flaws ahead of the final stage.
- Exposure to prototypes helps bring all the concepts together and lets stakeholders view the product from a fresh angle.

How to Create a Prototype? Tips and Best Practices

We now clearly understand **prototype design thinking**, its importance, and the several sorts of prototypes that can be used during the design process. The only thing left to do is to come up with your own. Here is some advice that will assist you in your journey:

1. Select the Appropriate Prototype

Before you start working on a prototype, think about where you are in the design process and the time and resources you have at your disposal. Early on, low-fidelity prototypes make sense, but as you release your product, you should switch to high-fidelity prototypes.

2. Set Up Clear Objectives

Know exactly what you expect your prototype to do before you begin. What, in other words, are you hoping to learn from testing your prototype? Always keep your research questions in mind as you concentrate on user demands.

3. Use Proper Equipment

If you're new to digital prototyping, experiment with some of the most well-liked tools available. Before you begin prototyping, choose a tool with the required features and capabilities and become familiar with its user interface. This will simplify the prototyping procedure.

4. Make a Move

Prototyping is a key component of design thinking. Therefore do it whenever you need to. As long as you keep in mind that the Design Thinking process isn't exactly linear, you can quickly create a prototype.

Tom Chi's approach to learning prototype

Tom Chi, formerly of Google, gives a [short but insightful talk](#) on how they prototyped early versions of [Google Glass](#) (kind of awesome website). He shares three rules for rapid prototyping:

1. Find the quickest path to experience.
2. Doing is the best kind of thinking.
3. Use materials that move at the speed of thought to maximize your rate of learning.

Stages in Prototyping

There are five design thinking prototyping phases

1. Define the Goal

The developer must create a broad vision for their product at this stage in the prototyping process. This stage could also function as a verbal description and could contain sketches. The creator can decide whether prototyping is necessary or whether they should reevaluate their original plan.

2. Highlight Core Elements

Attempting to make the prototype and final product similar is a mistake made by many creators. This is helpful when the prototype is used to convince investors of its value, but it's not required for most other uses. The designer should choose one or two standout aspects of their product to emphasize in their prototype.

3. Manufacture

The longest portion of the process is developing the prototype because the designer must weigh all the available possibilities. Consider using 3D printing, CNC machining, powder bed fusions, mold building, and casting as some common prototyping techniques. The decision will typically be influenced by the available materials, the schedule, and the price

4. Evaluation

After releasing the initial prototype, the designer will want to assess it, think about updating choices, and look for methods to streamline the procedure as a whole. This can call for a few minor adjustments, or it might mean throwing out the entire initial concept and beginning again at step one.

5. Presentation

Depending on the goal of the prototype, the presentation phase will change. It can entail producing many models to test with customers,

submitting the design for a patent, or showcasing it to potential investors. The presentation phase will assist in gauging interest and directing manufacturing techniques, depending on whether a straightforward small run or larger mass production is required.

Benefits of Prototyping in Design Thinking

Prototyping lets you enjoy the following benefits:

- Give all stakeholders a clear image of the potential benefits, risks, and costs connected with where a prototype might lead by having a solid basis to generate ideas for improvements.
- Helps foster a feeling of ownership in all interested parties, encouraging emotional commitment to the final success of the product.
- Reduces the number of mistakes that need to be fixed before the product is released to shorten time-to-market.
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Importance of a Prototype

Prototype is the preliminary version of the actual product developed to:

- Validate the design of the product,
- Present to investors or licensees,
- Protect the Intellectual property,
- Remove kinks in manufacturing,
- Test and refine the final product.

Qualities Of A Good Prototype

Generally, a prototype is considered to be a complete prototype only if it possesses these following characteristics:

- **Representation:** A prototype is a rudimentary representation of the actual product. It represents how the product will look and/or work like.
- **Precision:** More precise the prototype, better the response and feedback.
- **Functional:** A good prototype performs the basic functions of the actual product (if possible).

- **Improvisation:** A good prototype is one which can be improvised on with minimum effort. This one of the most important aspect of prototyping as a prototype is subject to many improvisations.

Types of Prototypes

Different project requirements require different types of prototypes. Such prototypes can differ on the basis of what they represent or how they are used.

Based On What They Represent

While a prototype can be in the form of paper, digital, miniature or a partial product, all of these can be categorized into three categories on the basis of what they represent.

1. Functional Prototypes

Functional prototypes are designed to imitate the functions of the actual product as closely as possible no matter how different they look from the actual product. These types of prototypes are produced for the products which are dependent on the function rather than the display.

For example: creating a backend prototype without working on the frontend of the website.

2. Display Prototypes

Display prototypes are designed with more focus on the look and feel of the product rather than the functions. These prototypes may or may not function but represent the look of the actual product very well.

Display prototypes are usually used in the fashion industry and in other industries where looks are more important.

3. Miniatures

Miniatures are smaller versions or the basic versions of the product focused on both the functional aspect as well as the display aspect. Nevertheless, these aren't the actual products and lack many qualities of the actual product like not working at full capacity, etc.

Miniatures are usually developed by the 3D printing of the product.

Based On How They Are Used

From a usability perspective, the prototypes can be categorized into:

1. Throwaway prototype

Throwaway prototype refers to the models which are eventually discarded or thrown away rather than becoming a part of the actual product. These products are only used to represent what an actual product can do. Throwaway prototypes are also called close ended prototypes

2. Evolutionary Prototype

Evolutionary prototyping uses a different approach than throwaway prototyping and involves building a basic but robust prototype in a manner which can further be improved and built upon to form an actual saleable product. This avoids wastage of resources.

Prototype Examples

Prototypes come in many types and shapes. It all depends on the reason for what a prototype is created. While some prototypes are developed just to represent or mimic the functioning or the look of the product (paper prototypes, HTML prototypes, etc.) to investors, some include showing a miniature version (3D print, single version of the lot, etc.) of the product with full or partial functionality.

Here are a few examples of prototypes:

Paper Prototype

A paper prototype is an example of a throwaway prototype created in the form of rough or hand-sketched drawings of the product's interface, front-end design, and sometimes the back end work.

3D print

The latest technology has made it possible to print a 3d version of your product which can actually work. But this technique is not feasible for mass production.

Digital Prototype

A digital prototype allows product developers to create a virtual model of the product which enables them to see how the individual components will work together and how the product will look once it's completed.

That is, it lets the developers virtually explore the complete product before it's actually built.

Scale Model

The scale model is a smaller and a non-functional model commonly used for prototyping large products like buildings, automobiles, etc.